MRR-Top0: A Topology-Aware Extension of Mean Reciprocal Rank for Semantic-Sensitive Retrieval Evaluation

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Abstract

Classical information retrieval metrics such as Mean Reciprocal Rank (MRR) evaluate only the position of the *first* relevant item, ignoring both deeper relevant results and the structural quality of the retrieved set within the corpus graph. We introduce **MRR-Top0**, a novel ranking metric that extends MRR to the entire top-k result list by weighting each relevant item's reciprocal rank with a *topology factor* $T_{q,i}$. This factor combines three graph signals computed on the corpus feature graph: *Personalized PageRank* (random-walk affinity from the query anchor), a *conductance penalty* (subgraph cohesion), and a *modularity gain* (community alignment). The resulting score evaluates both relevance order and structural coherence in a single, bounded, label-agnostic quantity. MRR-Top0 is applicable to any retrieval system operating over an embedding space equipped with a graph Laplacian—including spectral vector databases, RAG pipelines, and topology-aware search engines—and provides a computationally cheap proxy for assessing whether a ranking reflects the learned manifold structure of the dataset, not merely geometric proximity. We present the formal definition, explain every constituent, discuss key properties and practical guidance, and situate the metric within the broader landscape of graph-aware retrieval evaluation.

Keywords: information retrieval, evaluation metrics, graph topology, Personalized PageRank, conductance, modularity, spectral vector search, RAG, MRR

1 Introduction and Context of Applications

1.1 Limitations of Classical Retrieval Metrics

Modern AI systems, particularly Retrieval-Augmented Generation (RAG) pipelines, depend critically on the quality of their retrieval stage. A flawed retrieval step propagates errors downstream; as noted in recent enterprise analyses, "pure RAG is not really giving the optimal results that were expected" (1; 2). The metrics traditionally used to evaluate retrieval—Recall@k, Precision@k, MRR, NDCG@k—operate purely on rank positions and relevance labels. They are *semantics-blind* and *topology-blind*: they cannot distinguish a top-k set that forms a tight, well-connected cluster on the corpus manifold from one that is topologically scattered, even when both sets contain the same relevant items at the same ranks. These results in a low quality of tails and a low context-awareness of the results, this is particularly critical in vector space where semantic relevance of items is a key factor, like Retrieval Augmented Generation (RAG) pipelines and any other application related to Large Language Models (LLMs).

MRR in particular suffers from a narrow focus:

$$\mathrm{MRR} = \frac{1}{|Q|} \sum_{q=1}^{|Q|} \frac{1}{\mathrm{rank}(q, i_q^*)} \tag{1}$$

where i_q^* is the *first* relevant item for query q. This ignores all relevant items beyond rank 1 and says nothing about how the result set relates to the graph structure of the dataset.

NDCG@k partially addresses the multi-item issue via graded relevance and logarithmic discounting, but it still requires explicit relevance labels, is sensitive to shallow or incomplete pooling judgments, and has no notion of graph topology, community structure, or manifold coherence (3).

1.2 The Case for Topology-Aware Evaluation

When a vector database is backed by a graph Laplacian—as in spectral search systems such as ArrowSpace (4)—the corpus carries rich structural information: communities, random-walk affinities, conductance boundaries, and spectral signatures. A retrieval metric that ignores this structure leaves significant diagnostic power on the table.

Concretely, topology-awareness enables the following:

- **Drift detection:** A result set with high geometric similarity but low subgraph cohesion (high conductance) signals that the query is being answered from a topologically unstable region—a leading indicator of context drift and downstream hallucination.
- **Diversity assessment:** Modularity gain quantifies whether retrieved items span distinct communities or cluster into a single mode, supporting diversity-aware ranking.
- **Label-free evaluation:** In many production settings, graded relevance labels are unavailable or prohibitively expensive to maintain. Graph topology provides a complementary, label-agnostic signal.
- **RAG tail stability:** The tail of the top-k list is often where hallucinations originate. A metric that scores the entire list, weighted by structural quality, directly addresses this concern.

1.3 Target Application Domains

MRR-Top0 is designed for any system where retrieval quality interacts with graph structure. Primary application domains include:

1. **Spectral vector databases** (e.g., ArrowSpace, Genefold): indices that store a graph Laplacian alongside embeddings and use spectral scores (λ-mode, Rayleigh quotients) for similarity.
2. **RAG pipelines:** evaluating whether the context window fed to an LLM is structurally coherent, not just geometrically proximate.
3. **Knowledge-graph-augmented search:** systems that overlay a knowledge graph on an embedding space and route queries through graph neighborhoods.
4. **MLOps and dataset monitoring:** detecting distributional shift by tracking how MRR-Top0 evolves over time for a fixed query set—a drop in topology-factor scores signals that the manifold structure has changed.
5. **Mechanistic interpretability:** scoring feature circuits in sparse autoencoders (SAEs) by treating SAE decoder weights as items in a feature graph and using MRR-Top0 to evaluate retrieval of structurally coherent feature neighborhoods.

2 Formal Definition and Components

2.1 The MRR-Top0 Formula

Definition 1 (MRR-Top0)**.** *Let Q be a set of queries. For each query $q \in Q$, let* $\mathrm{Rel}(q) \subseteq \{1, \ldots, k\}$ *be the set of relevant items in the top-k result list. For each relevant item $i \in \mathrm{Rel}(q)$, let* $\mathrm{rank}(q, i)$ *denote its rank position and $T_{q,i}$ its* topology factor*. Then:*

$$\boxed{\text{MRR-Top0} = \frac{1}{|Q|} \sum_{q=1}^{|Q|} \sum_{i \in \mathrm{Rel}(q)} \frac{T_{q,i}}{\mathrm{rank}(q, i)}} \tag{2}$$

2.2 Explanation of Every Member

Table 1 provides a complete reference for each symbol in Equation (2).

Table 1: Members of the MRR-Top0 equation.

Symbol	Name	Description
$\lvert Q \rvert$	Query set size	The total number of evaluation queries. The outer $\frac{1}{\lvert Q \rvert}$ averages the per-query scores, exactly as in standard MRR.
q	Query index	An index running from 1 to $\lvert Q \rvert$, identifying each individual query in the evaluation set.
$\mathrm{Rel}(q)$	Relevant item set	The set of items judged relevant (or, in the label-agnostic regime, identified as relevant by the graph topology) for query q within the top-k cut-off. Unlike standard MRR which considers only the first relevant item, MRR-Top0 sums over *all* relevant items in $\mathrm{Rel}(q)$.
i	Item index	An index identifying a specific relevant item within the result list for query q.
$\mathrm{rank}(q, i)$	Rank position	The integer position (1-indexed) at which item i appears in the ranked result list for query q. The reciprocal $\frac{1}{\mathrm{rank}(q,i)}$ gives higher credit to items ranked higher—rank 1 contributes 1.0, rank 5 contributes 0.2, etc.
$T_{q,i}$	Topology factor	A bounded scalar $T_{q,i} \in [0, 1]$ (practically $\approx [0.2, 1.0]$ with well-tuned weights) that modulates the reciprocal-rank contribution of item i based on its structural quality within the result sub-graph. Defined in §2.3. This is the core novelty of MRR-Top0.

2.3 The Topology Factor

The topology factor $T_{q,i}$ combines three graph signals to reward items that are structurally well-positioned relative to the query and the accumulated result set:

Definition 2 (Topology Factor)**.**

$$T_{q,i} = \alpha_1 \cdot \mathrm{PPR}(q,i) + \alpha_2 \cdot \big(1 - \phi(S_{q,1:i})\big) + \alpha_3 \cdot \mathcal{Q}(S_{q,1:i}) \tag{3}$$

where $\alpha_1 + \alpha_2 + \alpha_3 = 1$ *and* $\alpha_j > 0$.

Each component is explained in detail below.

2.3.1 Personalized PageRank: $\mathrm{PPR}(q,i)$

Personalized PageRank (PPR) (5) computes the stationary distribution of a random walk that, at each step, either follows an outgoing edge with probability d (the damping factor, typically 0.85) or teleports back to the query node q with probability $1-d$. The resulting score $\mathrm{PPR}(q,i)$ measures the *random-walk affinity* from the query anchor q to item i on the corpus feature graph:

$$\mathbf{p}_q = (1-d)\,\mathbf{e}_q + d\,\mathbf{M}^\top \mathbf{p}_q \tag{4}$$

where $\mathbf{e}_q$ is a one-hot vector at the query node and $\mathbf{M}$ is the column-stochastic transition matrix of the feature graph. The i-th entry of $\mathbf{p}_q$ gives $\mathrm{PPR}(q,i)$.

Role in $T_{q,i}$. PPR ensures that the topology signal is *query-anchored*: items that are geometrically similar to q (high cosine) but poorly connected via the feature graph (low PPR) receive a reduced topology factor. This prevents "shortcuts" where a distant item happens to match the query embedding but resides in a different topological community.

2.3.2 Conductance Penalty: $1 - \phi(S_{q,1:i})$

Let $S_{q,1:i}$ denote the partial result set comprising the top i items for query q. The *conductance* of this set is:

$$\phi(S_{q,1:i}) = \frac{|\partial(S_{q,1:i})|}{\min\big(\mathrm{vol}(S_{q,1:i}),\ \mathrm{vol}(V \setminus S_{q,1:i})\big)} \tag{5}$$

where $\partial(S)$ is the set of edges crossing the boundary of S, and $\mathrm{vol}(S) = \sum_{v \in S} \deg(v)$ is the volume (sum of degrees) of S (6).

Role in $T_{q,i}$. Low conductance ($\phi \approx 0$) indicates a tightly connected subgraph with few boundary edges—a cohesive, topologically stable result set. High conductance ($\phi \to 1$) signals a loose, scattered subgraph. The term $1-\phi$ therefore *rewards* cohesion and *penalizes* topological scatter. Crucially, this is evaluated *incrementally*: $\phi(S_{q,1:i})$ is the conductance of the result set accumulated up to rank i, so the topology factor captures how each new item affects subgraph quality.

2.3.3 Modularity Gain: $\mathcal{Q}(S_{q,1:i})$

Modularity measures the fraction of intra-community edges in the result subgraph compared to a random null model:

$$\mathcal{Q}(S_{q,1:i}) = \frac{1}{2m} \sum_{u,v \in S_{q,1:i}} \left[A_{uv} - \frac{\deg(u)\deg(v)}{2m} \right] \tag{6}$$

where A is the adjacency matrix of the corpus graph and $m = \frac{1}{2}\sum_{uv} A_{uv}$ is the total number of edges (6).

Role in $T_{q,i}$. High modularity indicates that the result set respects the intrinsic community structure of the corpus—items belong to the same or closely related clusters. Low modularity (or negative values) indicates that results are drawn from across unrelated communities, suggesting potential topical incoherence.

2.3.4 Weight Parameters: $\alpha_1, \alpha_2, \alpha_3$

The weights $\alpha_1, \alpha_2, \alpha_3$ control the relative influence of each graph signal and must sum to 1. Recommended defaults for query-centric retrieval are:

$$\alpha_1 = 0.5, \quad \alpha_2 = 0.3, \quad \alpha_3 = 0.2 \tag{7}$$

This weighting prioritizes query-anchored affinity (PPR) while giving secondary importance to subgraph cohesion and tertiary importance to community alignment. The weights are tunable per task: for diversity-heavy applications, increasing α_3 rewards cross-community coverage.

2.4 Practical Guidance

1. **Set k cutoff.** Limit $\mathrm{Rel}(q)$ to top-k (e.g., $k = 10$) to match user attention and reduce computational cost of topology metrics.
2. **Normalize $T_{q,i}$.** Scale PPR, conductance, and modularity to $[0, 1]$ so the topology factor is bounded and interpretable. Typical values: $T_{q,i} \in [0.2, 1.0]$.
3. **Graph construction.** Build a k-NN or radius graph on corpus embeddings (e.g., $k = 15$, cosine similarity) and precompute community structure. Update topology factors incrementally as results are ranked.
4. **Comparison baseline.** Report both MRR (geometry-only) and MRR-Top0 (geometry + topology) to quantify the added value of structural signals. Improvements of 5–15% in MRR-Top0 indicate meaningful topology contribution.

2.5 Key Properties and Qualities

Proposition 1 (Generalization of MRR)**.** *When $T_{q,i} = 1$ for all i and only the first relevant item is considered, MRR-Top0 reduces to standard MRR (Equation 1).*

Proof. Setting $T_{q,i} = 1$ and $|\mathrm{Rel}(q)| = 1$ with $\mathrm{Rel}(q) = \{i_q^*\}$, Equation (2) becomes:

$$\text{MRR-Top0} = \frac{1}{|Q|} \sum_{q=1}^{|Q|} \frac{1}{\mathrm{rank}(q, i_q^*)} = \text{MRR}.$$

□

The following properties make MRR-Top0 uniquely suited for modern retrieval evaluation:

1. Multi-rank evaluation. MRR-Top0 aggregates contributions from *all* relevant items in the top-k, not just the first hit. This addresses MRR's well-known limitation of ignoring deeper relevant results, which is particularly problematic in RAG systems where the LLM consumes the entire context window.

2. Topology-awareness. By weighting reciprocal ranks with PPR, conductance, and modularity, MRR-Top0 scores how well the ranking reflects both relevance and graph structure. This avoids sole reliance on geometric similarity, which is known to produce topologically scattered result sets under domain shift.

Table 2: Comparison of MRR-Top0 with standard metrics.

Property	**MRR**	**NDCG@k**	**Recall@k**	**MRR-Top0**
Evaluates full top-k	×	✓	✓	✓
Rank-aware	✓	✓	×	✓
Topology-aware	×	×	×	✓
Label-agnostic mode	×	×	×	✓
Bounded output	$[0, 1]$	$[0, 1]$	$[0, 1]$	$[0, H_k]$
Penalizes scattered results	×	×	×	✓
Community alignment	×	×	×	✓
Robust to incomplete labels	N/A	×	×	✓

3. Label-agnostic operation. MRR-Top0 can operate without explicit graded relevance judgments. When labels are unavailable, $\mathrm{Rel}(q)$ can be defined via graph-based criteria (e.g., items within a PPR threshold of q), and the topology factor provides the quality signal. This makes the metric robust to the pooling bias and shallow judgment problems that degrade NDCG reliability.

4. Harmonic mean interpretation. The reciprocal of MRR-Top0 corresponds to a topology-weighted harmonic mean of relevant ranks, analogous to MRR's harmonic mean of first-hit positions. This preserves the intuitive interpretation: a score close to 1.0 means relevant, structurally well-positioned items appear near the top of the list.

5. Incremental subgraph quality. The conductance and modularity components are evaluated on the *partial* result set $S_{q,1:i}$, not just on individual items. This means MRR-Top0 captures how each successive item affects the overall structural quality of the retrieved set—a feature no existing single-number IR metric provides.

6. Computationally efficient. PPR can be approximated in $O(1/\epsilon)$ time per query via bidirectional estimators (5). Conductance and modularity are computed incrementally on the growing subgraph. The overhead relative to standard MRR is modest, making MRR-Top0 practical for production benchmarking.

2.6 Worked Example

Suppose query q retrieves results at ranks 1, 3, 5 with relevance judgments $\{1, 3, 5\} = \mathrm{Rel}(q)$ and topology factors $T_{q,1} = 0.9$, $T_{q,3} = 0.7$, $T_{q,5} = 0.5$:

$$\begin{aligned} \text{MRR-Top0}(q) &= \frac{0.9}{1} + \frac{0.7}{3} + \frac{0.5}{5} \\ &= 0.9 + 0.233 + 0.1 \\ &= 1.233 \end{aligned} \tag{8}$$

Standard MRR would give $\frac{1}{1} = 1.0$, ignoring ranks 3 and 5 and all topology. MRR-Top0's higher per-query score reflects credit for (a) multiple relevant items beyond the first hit and (b) their structural quality as measured by the topology factor. When averaged over $|Q|$ queries, the metric remains interpretable as a mean topology-weighted reciprocal rank.

2.7 Reference Implementation

Algorithm 1 MRR-Top0 for a single query

Require: Retrieved list R, relevance dict, topology factors T, cutoff k
Ensure: MRR-Top0 score for query q

```
score ← 0.0
for (i, rank) in R do
    if rank > k then break
    end if
    if relevance[i] > 0 then
        score ← score + T[i] / rank
    end if
end for
return score
```

The full MRR-Top0 over a query set is the mean of per-query scores. A Python reference implementation is available in the ArrowSpace test suite (test_15_CVE_pagerank.py).

2.8 MRR-Top0 and Epiplexity.

Topological PageRank and epiplexity share a common mathematical substrate: the graph Laplacian and its spectral decomposition. Classical PageRank computes the stationary distribution of a random walk on a graph, and Chung (11) established that PageRank vectors are expressible through the combinatorial Laplacian $L = D - A$ and its discrete Green's function, with the Fiedler eigenvalue λ_1 governing the rate of convergence of the walk. Shur et al. (10) further proved that log-PageRank embeddings converge to the Laplacian eigenvectors for regular graphs, making personalised PageRank a locally computable proxy for spectral position. **Epiplexity (12), defined as the structural information extractable by a computationally bounded observer**, measures the area under the loss curve above the final loss—equivalently, the information in the program that minimises the time-bounded minimum description length. Items with high structural content relative to the learned manifold induce richer internal representations (more reusable circuits), and this property correlates with out-of-distribution generalisation. The connection is direct: personalised PageRank $\mathrm{PPR}(q, i)$ in MRR-Top0's topology factor $T_{q,i}$ measures random-walk affinity on the same graph whose Laplacian spectrum encodes the structural complexity that epiplexity formalises. High-PPR items are those reachable via smooth, low-frequency paths on the manifold—the same regime where epiplexity predicts high structural information content—while low-PPR items occupy spectrally rough, high-frequency regions where the observer extracts less transferable structure. In this sense, MRR-Top0's topology-weighted reciprocal rank can be interpreted as an *epiplexity-aware* retrieval score: it rewards not merely geometric proximity but manifold-consistent structural alignment.

ArrowSpace makes the synergy between MRR-Top0 and epiplexity operationally realisable by providing the λ_i (Rayleigh quotient of each item against $L = \mathrm{Laplacian}(C^\top)$, the feature-space graph Laplacian over centroids) as a precomputed, $O(1)$- per-item scalar. Even if the epiplexity interpretation of λ is approximate, this value serves as a computationally cheap proxy for "how much an item deviates from learned structure," which is operationally useful for active learning and OOD detection in retrieval systems. In practice, the personalised PageRank component of $T_{q,i}$ can be replaced or augmented by spectral proximity $\exp(-|\lambda_q - \lambda_i|/\sigma_\lambda)$, yielding a fully spectral MRR-Top0 variant that requires no explicit random-walk computation—only the already-indexed λ values and the τ-mode search that ArrowSpace provides at query time.

3 Conclusion

MRR-Top0 was developed as part of the ArrowSpace spectral vector database project (13) (`cargo add arrowspace`, `pip install arrowspace`). The metric design builds on the manifold invariant $\mathbf{L} = \text{Laplacian}(\text{Transpose}(\mathbf{C}))$, where $\mathbf{C}$ is the centroid matrix, ensuring the topology is computed in *feature-space* rather than items-space. The epiplexity interpretation of the Rayleigh quotient λ_i as a spectral deviation proxy provides the operational grounding: even when approximate, λ offers a computationally cheap signal for how much an item deviates from learned structure, which is operationally useful for active learning and OOD detection in RAG systems. A complete script is available in this repository.

The integration of MRR-Top0 with the spectral infrastructure of ArrowSpace represents a fundamental advancement in quantifying search quality by grounding retrieval evaluation in the manifold geometry of the data. While traditional metrics like NDCG or classic MRR remain blind to the structural coherence of the result set, MRR-Top0's topology factor $T_{q,i}$ explicitly rewards items that exhibit high random-walk affinity and community alignment. This transition from purely geometric proximity to topological consistency is made computationally feasible through the extreme compression of the graph Laplacian into ArrowSpace's scalar Rayleigh quotients (λ). By storing only these spectral signatures rather than the full adjacency matrix, the system reduces the complex global dynamics of a random walk to a series of local, $O(1)$ similarity checks in λ-space.

This synergy allows for a new class of "epiplexity-aware" retrieval systems that can detect out-of-distribution (OOD) drift and measure the structural information content of a query in real-time. Because the PageRank vectors of a graph converge to the eigenvectors of its Laplacian, the λ values provided by ArrowSpace serve as a precomputed proxy for an item's stationary position within the manifold's energy landscape. Consequently, the advancement brought by MRR-Top0 is not merely a more sensitive ranking score, but a unified framework where the quality of a search result is intrinsically linked to its ability to preserve the learned structure of the feature space, ensuring robust and stable performance across the long tail of enterprise datasets.

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